

Lecture 7:

Error Estimation for Classification

Spring 2026

10 ECTS Credits

Department of Information and Communication Technology

Faculty of Engineering and Science

Recap

- Rosenblatt's perceptron.
- Support Vector Machines
 - Linear SVM on linearly separable data.
 - General Linear SVMs
 - Non-linear SVMs
- Neural Networks:
 - Fully connected NN
 - Nonlinearities for NN
 - NN Classifier
 - Loss functions for NN
 - Backpropagation
 - Enablers of deep NN:
 - Non-saturating non-linearities
 - Weight Initialization
 - Stochastic Gradient Descent (SGD)
 - Learning rate scheduling
- Decision Trees
 - CART Rule for Decision Trees



Error Estimation for Classification

- We don't know the feature-label distribution of a problem in most practical cases.
- Computing the exact error of a classifier is impossible.
- With large sample size, we could:
 - Divide dataset into train and test sets,
 - Design the classifier on training set,
 - Evaluate on test set.
- In abundance of test data, the above method produces an accurate error estimator.
- Large test sets are not always available/possible.

Error Estimation Rules

- What is a good classification rule?
- A good classification rule produces classifiers with small average error rates.
- What if our error rate estimation is not accurate? What if the error rate cannot be stated with confidence?
- Goodness of a classification rule can only be quantified with a **good error estimation rule**.
- Even though Bayes error ε^* and expected error rate μ_n can be estimated, focus is on ε_n , error of a sample-based classifier.

Error Estimation for Classification

- A classification rule Ψ_n and sample data $S_n = \{(\mathbf{X}_1, Y_1), (\mathbf{X}_2, Y_2), \dots, (\mathbf{X}_n, Y_n)\}$
- Error estimation rule is the mapping $\Xi_n: (\Psi_n, S_n, \xi) \mapsto \hat{\varepsilon}_n$, where $0 \leq \hat{\varepsilon}_n \leq 1$
- ξ represents any internal random factors of Ξ_n , independent on the random sample data.
- If there are no ξ , error estimation is **non-randomized**, otherwise **randomized**.
- With Ψ_n, S_n and ξ fixed, $\hat{\varepsilon}_n$ is an error estimate of the classifier $\psi_n = \Psi_n(S_n)$.
- With only Ψ_n , $\hat{\varepsilon}_n$ is an error estimator.

Error Estimation for Classification

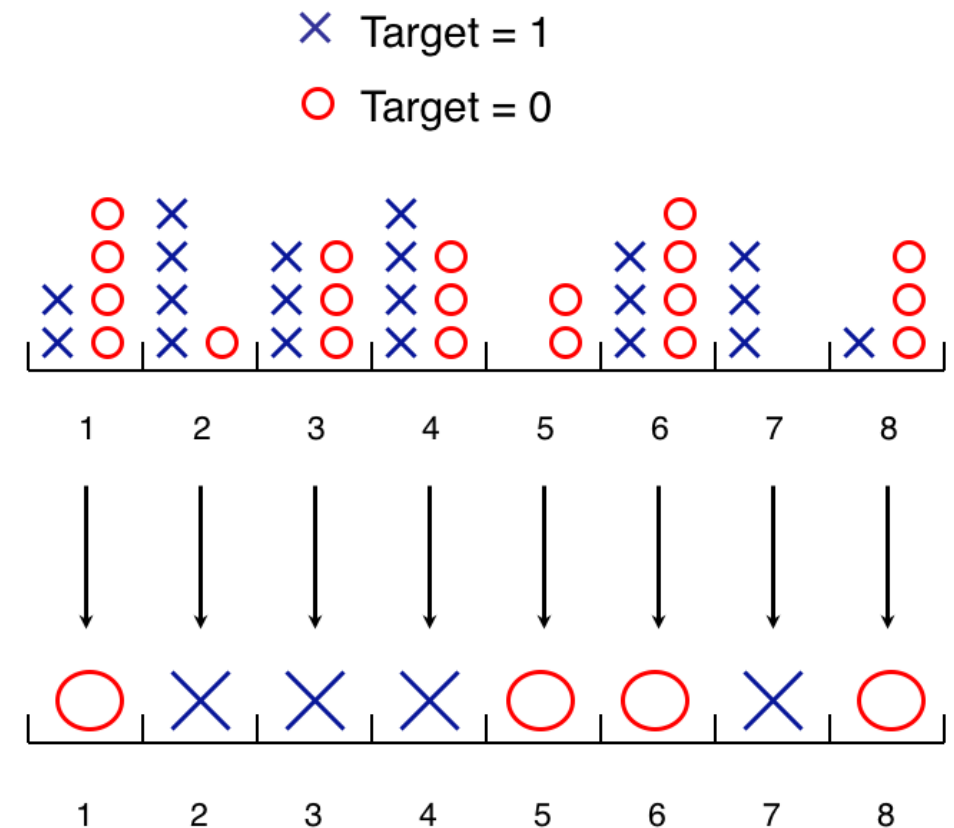
- While an error estimation rule is a general procedure, an error estimator is associated with a classification rule.
- Error estimators can have different properties, and performance can vary with different classification rules.
- The goodness of a classification procedure w.r.t. a specific feature-label distribution involves checking:
 - how well ε_n approximates ε^* , and
 - how well $\hat{\varepsilon}_n$ approximates ε_n .
- The latter depends on the joint distribution between $\hat{\varepsilon}_n$ and ε_n .

Resubstitution Error Estimation Rule

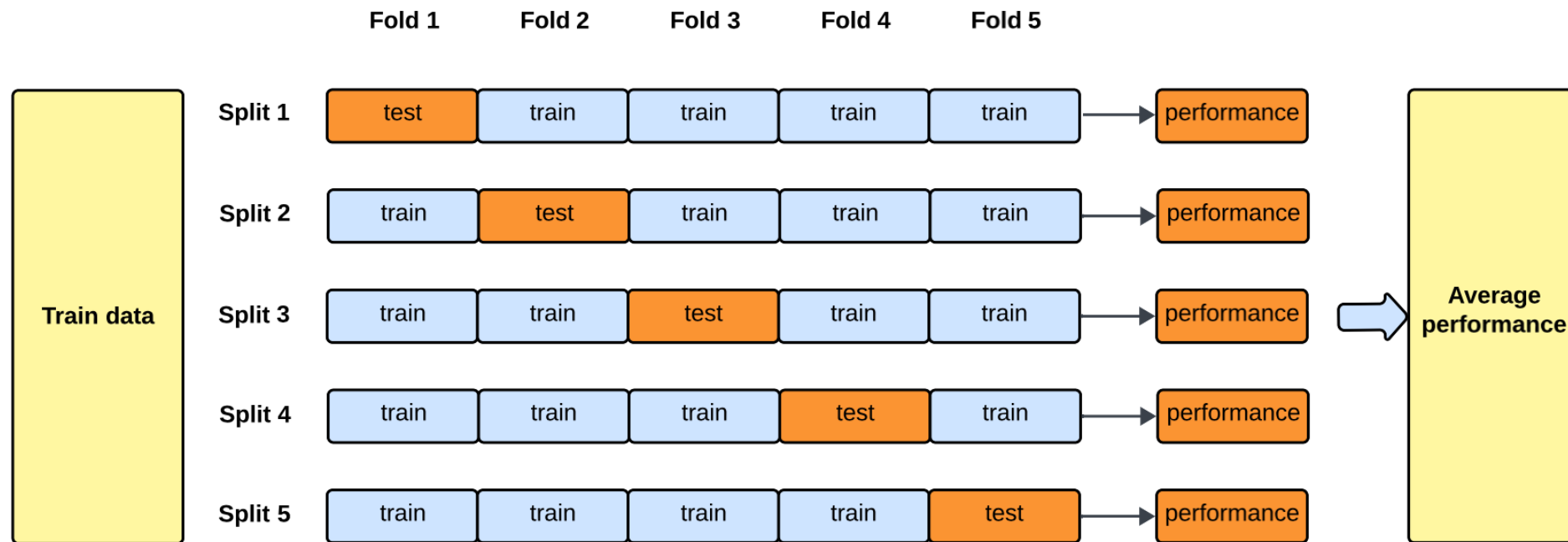
- Based on testing the classifier on the training data.

$$\mathbb{E}_n(\Psi_n, S_n) = \frac{1}{n} \sum_{i=1}^n |Y_i - \Psi_n(S_n)(\mathbf{X}_i)|$$

- Also known as **apparent error**, **training error**, **empirical error**.
- Fraction of errors on S_n by the classifier designed on S_n .
- Since this is a non-randomized error estimation rule, ξ is omitted.



Cross-Validation Error Estimation Rule



Cross-Validation Error Estimation Rule

- In k -fold cross-validation, the sample is partitioned at random into k equal sized folds:

$$S_{(j)} = \left\{ \left(\mathbf{X}_i^{(j)}, Y_i^{(j)} \right), \quad \forall i = 1, \dots, n/k \right\}, \quad \forall j = 1, \dots, k$$

- Each fold $S_{(j)}$ is left out during design process to obtain a deleted sample $S_n - S_{(j)}$.
- A classification rule $\Psi_{n-n/k}$ is applied to $S_n - S_{(j)}$ and the resulting error is calculated on $S_{(j)}$.
- The average of these k error rate is the cross-validated error estimate:

$$\mathbb{E}_n(\Psi_n, S_n, \xi) = \frac{1}{n} \sum_{j=1}^k \sum_{i=1}^{n/k} \left| Y_i^{(j)} - \Psi_{n-n/k}(S_n - S_{(j)})(\mathbf{X}_i^{(j)}) \right|.$$

Cross-Validation Error Estimation Rule

$$\Xi_n(\Psi_n, S_n, \xi) = \frac{1}{n} \sum_{j=1}^k \sum_{i=1}^{n/k} \left| Y_i^{(j)} - \Psi_{n-n/k}(S_n - S_{(j)})(\mathbf{X}_i^j) \right|$$

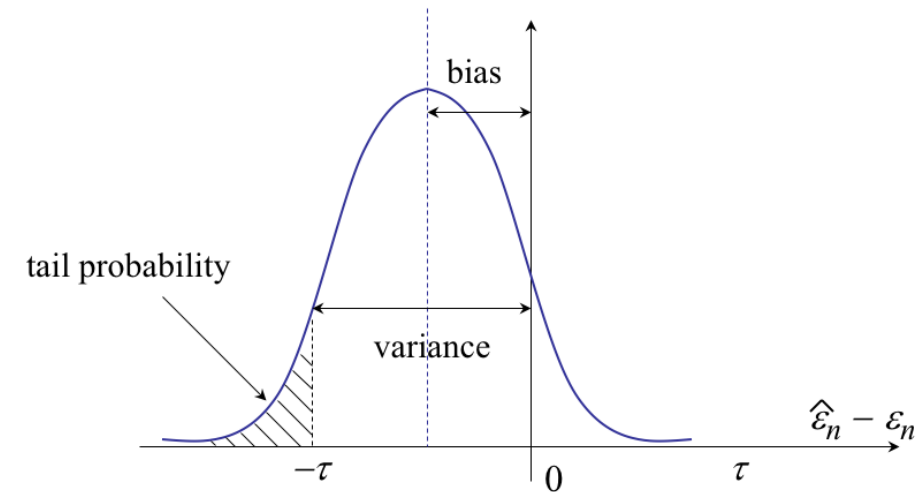
- The randomness in partitioning the sample S_n into folds is specified by ξ .
- The general case of k -fold cross-validation is a randomized error estimation rule.
- What if $k = n$? Or when each fold contains a single point.
- Randomness disappears.
- Also known as **leave-one-out error estimation rule**.

Error Estimation Performance

- Depends on:
 - Classification rule used,
 - Feature-label distribution, and
 - Sample size.
- We are interested in the «average» performance of the error estimator over all possible sample data.
- We are going to look at the performance criteria for error estimators:
 - Deviation with respect to true error,
 - Bias, variance, RMS, and tail probabilities, and
 - Consistency

Deviation Distribution

- The purpose of an error estimator is to approximate the true error ε_n .
- How the estimated error $\hat{\varepsilon}_n$ deviates from the true error ε_n is significant to discuss the accuracy of the error estimator.
- The figure shows the distribution of the **deviation** $\hat{\varepsilon}_n - \varepsilon_n$.
- As opposed to a joint distribution, this is univariate and contains a useful subset of the information.



Bias, Variance, RMS, and Tail Probabilities

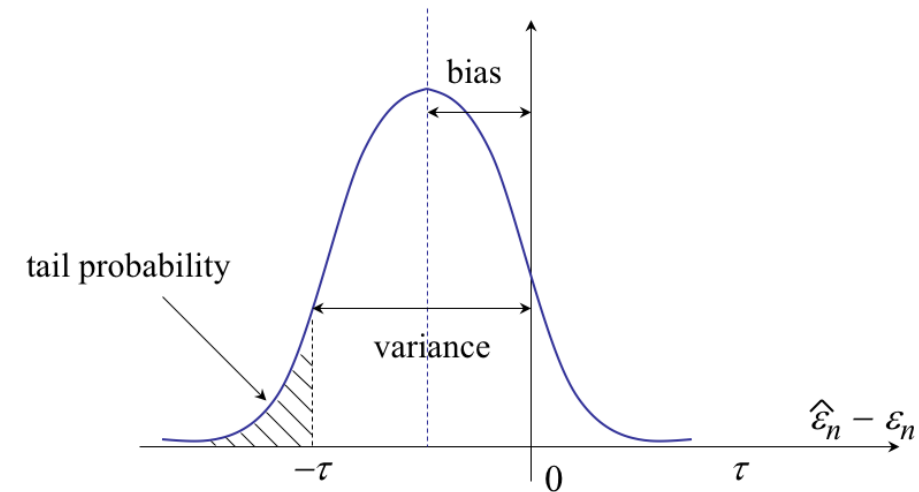
- From the deviation distribution, we derive four key performance metrics:

- Bias:**

- $\text{Bias}(\hat{\varepsilon}_n) = E[\hat{\varepsilon}_n - \varepsilon_n] = E[\hat{\varepsilon}_n] - E[\varepsilon_n]$
- Optimistically biased, if $\text{Bias}(\hat{\varepsilon}_n) < 0$.
- Pesimistically biased, if $\text{Bias}(\hat{\varepsilon}_n) > 0$.
- Unbiased, if $\text{Bias}(\hat{\varepsilon}_n) = 0$.

- Variance:**

- $\text{Var}_{dev}(\hat{\varepsilon}_n) = \text{Var}(\hat{\varepsilon}_n - \varepsilon_n) = \text{Var}(\hat{\varepsilon}_n) + \text{Var}(\varepsilon_n) - 2\text{Cov}(\hat{\varepsilon}_n, \varepsilon_n)$
- Measures how individual error estimates typically differ from the mean.



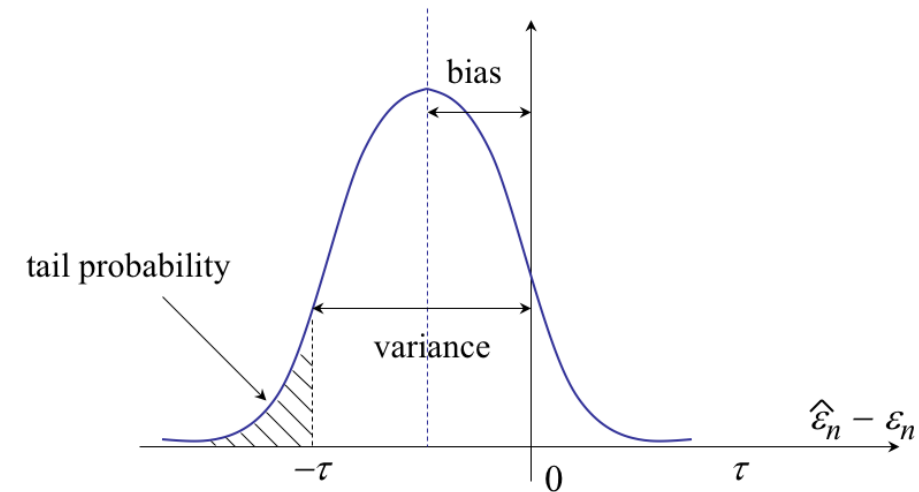
Bias, Variance, RMS, and Tail Probabilities

- **RMS:**

- $\text{RMS}(\hat{\varepsilon}_n) = \sqrt{E[(\hat{\varepsilon}_n - \varepsilon_n)^2]} = \sqrt{\text{Bias}(\hat{\varepsilon}_n)^2 + \text{Var}_{dev}(\hat{\varepsilon}_n)}$
- A comprehensive measure of error estimation accuracy

- **Tail Probabilities:**

- Quantify the likelihood of observing extreme deviations between $\hat{\varepsilon}_n$ and ε_n above a threshold τ .
 - Given by $P(|\hat{\varepsilon}_n - \varepsilon_n| \geq \tau)$.
 - Important in establishing confidence intervals and assessing worst case scenarios.
- A good error estimation performance require all these as close to zero as possible.



Randomized vs. Nonrandomized Error Estimators

- If the error estimation rule is randomized, $\hat{\epsilon}_n$ is still random after S_n is specified.
- The **internal variance** of $\hat{\epsilon}_n$

$$\text{Var}_{int} = \text{Var}(\hat{\epsilon}_n | S_n)$$

- Var_{int} measures the variability due only to internal random factors and not the variability due to sample S_n and internal random factor is specified by ξ .
- $\text{Var}(\hat{\epsilon}_n)$ includes both variability due to internal random factors and due to the sample. We can write
$$\text{Var}(\hat{\epsilon}_n) = E[\text{Var}(\hat{\epsilon}_n | S_n)] + \text{Var}(E[\hat{\epsilon}_n | S_n]) = E[\text{Var}_{int}] + \text{Var}(E[\hat{\epsilon}_n | S_n])$$
- For non-randomized $\hat{\epsilon}_n$, $\text{Var}_{int} = 0$ and for randomized $\hat{\epsilon}_n$, $E[\text{Var}_{int}] > 0$.
- Randomized error estimation rules attempt to minimize internal variance by averaging and intensive computation.

Consistency

- An error estimator is:
 - **consistent** if as $n \rightarrow \infty$, $\hat{\varepsilon}_n \rightarrow \varepsilon_n$ in probability.
 - **strongly consistent** if the above convergence is with probability 1.

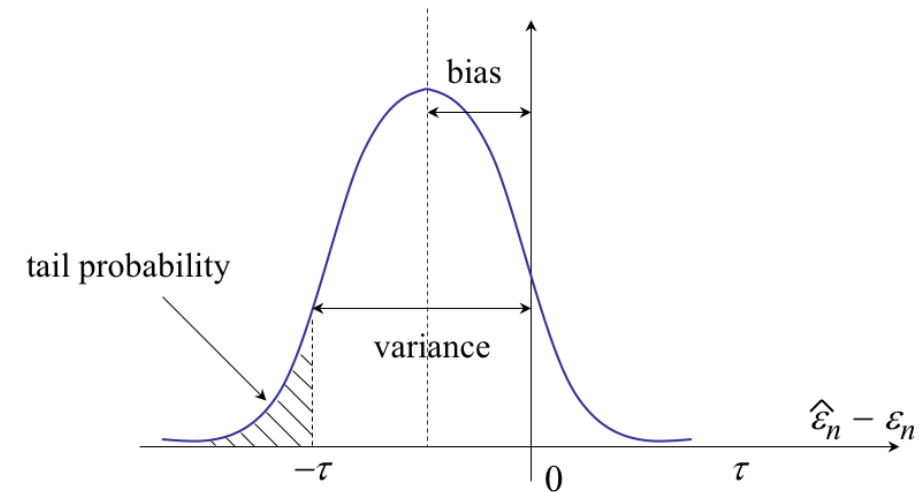
- Consistency is related to tail probabilities of the deviation distribution:

$$\lim_{n \rightarrow \infty} P(|\hat{\varepsilon}_n - \varepsilon_n| \geq \tau) = 0, \quad \forall \tau > 0$$

- A strong consistency demands the stronger condition:

$$\sum_{n=1}^{\infty} P(|\hat{\varepsilon}_n - \varepsilon_n| \geq \tau) < \infty, \quad \forall \tau > 0$$

- If an estimator is consistent regardless of the feature-label distribution, it is **universally consistent**.



Consistency

- If a classification rule is consistent, $\varepsilon_n \rightarrow \varepsilon^*$.
- If an error estimator is consistent, $\hat{\varepsilon}_n \rightarrow \varepsilon_n$.
- Thus, if both are consistent, $\hat{\varepsilon}_n \rightarrow \varepsilon^*$.
- If both the classification rule and error estimator are consistent, the error estimator provides a good estimate the Bayes error, with sufficiently large sample S_n .
- The rate of convergence of $\hat{\varepsilon}_n$ to ε_n can be bound for some classification rules and error estimators.
- There will always be distributions for which $\hat{\varepsilon}_n$ converges to ε^* arbitrarily slowly.
- Unless additional information is available about the feature-label distribution, it cannot be guaranteed that $\hat{\varepsilon}_n$ is close to ε^* for a given n .

Test-Set Error Estimation

- Assumes availability of a second random sample $S_m = \left\{ \left(\mathbf{X}_i^{(t)}, Y_i^{(t)} \right), \forall i = 1, \dots, m \right\}$, known as **test data**.
- This sample is independent and identically distributed with the training data S_n , and is not used in classifier design.
- The classification rule is applied on the training data and the error of the resulting classifier is estimated on the test data.
- The **test-set** or **holdout error estimation rule error** is formulated as:

$$\mathbb{E}_{n,m}(\Psi_n, S_n, S_m) = \frac{1}{m} \sum_{i=1}^m \left| Y_i^{(t)} - \psi_n \left(\mathbf{X}_i^{(t)} \right) \right|$$

- This is a randomized error estimation rule, with test data themselves as internal random factors, $\xi = S_m$.

Test-Set Error Estimation

- For any classification rule Ψ_n , the test-set error estimator is given by

$$\hat{\varepsilon}_{n,m} = \Xi_{n,m}(\Psi_n, S_n, S_m)$$

- The estimator has some interesting properties:
 - It is unbiased, if the test and train sets come from the same distribution:

$$E[\hat{\varepsilon}_{n,m} | S_n] = \varepsilon_n, \quad m = 1, 2, \dots$$

- Given the training data S_n , $\hat{\varepsilon}_{n,m} \rightarrow \varepsilon_n$ with probability 1 as $m \rightarrow \infty$, regardless of the feature-label distribution.
- In addition, from the above sense of unbiasedness, $E[\hat{\varepsilon}_{n,m}] = E[\varepsilon_n]$.
Hence:

$$\text{Bias}(\hat{\varepsilon}_{n,m}) = E[\hat{\varepsilon}_{n,m}] - E[\varepsilon_n] = 0$$

Test-Set Error Estimation

- Internal variance of the randomized estimator $\hat{\varepsilon}_{n,m}$ vanishes as $m \rightarrow \infty$ and is bounded:

$$\text{Var}_{int}(\hat{\varepsilon}_{n,m}) \leq \frac{1}{4m}$$

- There is a distribution-independent bound on the RMS:

$$\text{RMS}(\hat{\varepsilon}_{n,m}) \leq \frac{1}{2\sqrt{m}}$$

with 400 test points, RMS is guaranteed to be no larger than 2.5% regardless of classification rule and feature-label distribution.

- In practice, there will be n labelled sample points, which are split into training S_{n-m} and testing S_m sets.

Test-Set Error Estimation

- For large n , $E[\varepsilon_{n-m}] \approx E[\varepsilon_n]$ such that:
 - training set is still large enough (S_{n-m}) to train an accurate classifier,
 - test set is still large enough (S_m) to obtain a good test-set estimator.
- Splitting the sample into train and test means fewer data for training the classifier, resulting in poorer performing classifier.
- If fewer samples are set aside for test, there is insufficient data for proper testing.
- Test-set error estimator thus becomes infeasible when labelled data is expensive to obtain.

Redistribution Error Estimation

- Simplest, and fastest estimation rule.
- Based solely on training data.
- Given the classification rule Ψ_n and classifier $\psi_n = \Psi_n(S_n)$ designed on S_n :

$$\hat{\varepsilon}_n = \frac{1}{n} \sum_{i=1}^n |Y_i - \psi_n(\mathbf{X}_i)|$$

- Alternatively, redistribution error estimator can be defined as:

$$\hat{\varepsilon}_n = E_{p_n}[|Y - \psi_n(\mathbf{X})|]$$

- Using the **empirical feature-label distribution** as:

$$p_n(\mathbf{X}, Y) = P(X = X_i, Y = Y_i \mid S_n) = \frac{1}{n}, \quad \forall i = 1, \dots, n$$

Redistribution Error Estimation

- The problem with resubstitution error estimator is that it is typically optimistically biased: $\text{Bias}(\hat{\epsilon}_n) < 0$.
- It tends to underestimate the true classification error, as it evaluates on the same data that was used to train the model.
- Unacceptably large bias for overfitting classification rule, especially in small-sample situations.
- Example: 1-nearest neighbour classification rule.

Cross-Validation Error Estimation

- Improves on the bias of resubstitution by using a resampling strategy.
- Classifiers are trained and tested on non-overlapping subsets of data.
- Basic k -fold cross-validation was discussed earlier.
- In **stratified cross-validation**, to reduce variance, the classes are represented in each fold in the same proportion as original data.
- To limit the effect of internal variance, the process of random selection of folds can be repeated several times and the results can be averaged resulting in **repeated k -fold cross validation**.
- **Complete k -fold cross validation** is the limit case where all possible folds of size n/k are used, eliminating internal variance.

Cross-Validation Error Estimation

- The k -fold cross-validation error estimator has **near-unbiasedness property**:

$$E[\hat{\varepsilon}_n] = E[\varepsilon_{n-n/k}]$$

- The above distribution-independent property holds as long as sampling is independently and identically distributed.
- However, the k -fold cross-validation error estimator is pessimistically biased (in estimating the true error rate on entire dataset):

$$E[\varepsilon_{n-n/k}] > E[\varepsilon_n]$$

- $E[\varepsilon_{n-n/k}]$ is based on smaller sample size than $E[\varepsilon_n]$.
- Increasing the number of folds can solve the bias problem to some extent, however, this increases the estimator variance.
- The maximum value for $k = n$, which corresponds to leave-one-out error estimator, where $E[\hat{\varepsilon}_n^\ell] = E[\varepsilon_{n-1}]$

Cross-Validation Error Estimation

- The major concern regarding cross-validation is not bias but variance.
- Even though the leave-one-out cross-validation is a non-randomized error estimation rule, it can exhibit large variance (small test sample size $n/k = 1$).
- The variance could be understood as it attempts to estimate the error of a classifier $\psi_n = \Psi_n(S_n)$ by estimating the error of surrogate classifiers $\psi_{n,i} = \Psi_{n-n/k}(S_n - S_{(i)})$, $\forall i = 1, \dots, k$.
- If the classification rule is overfitting, deleting different points from sample produces wildly different surrogate classifiers.

Cross-Validation Error Estimation

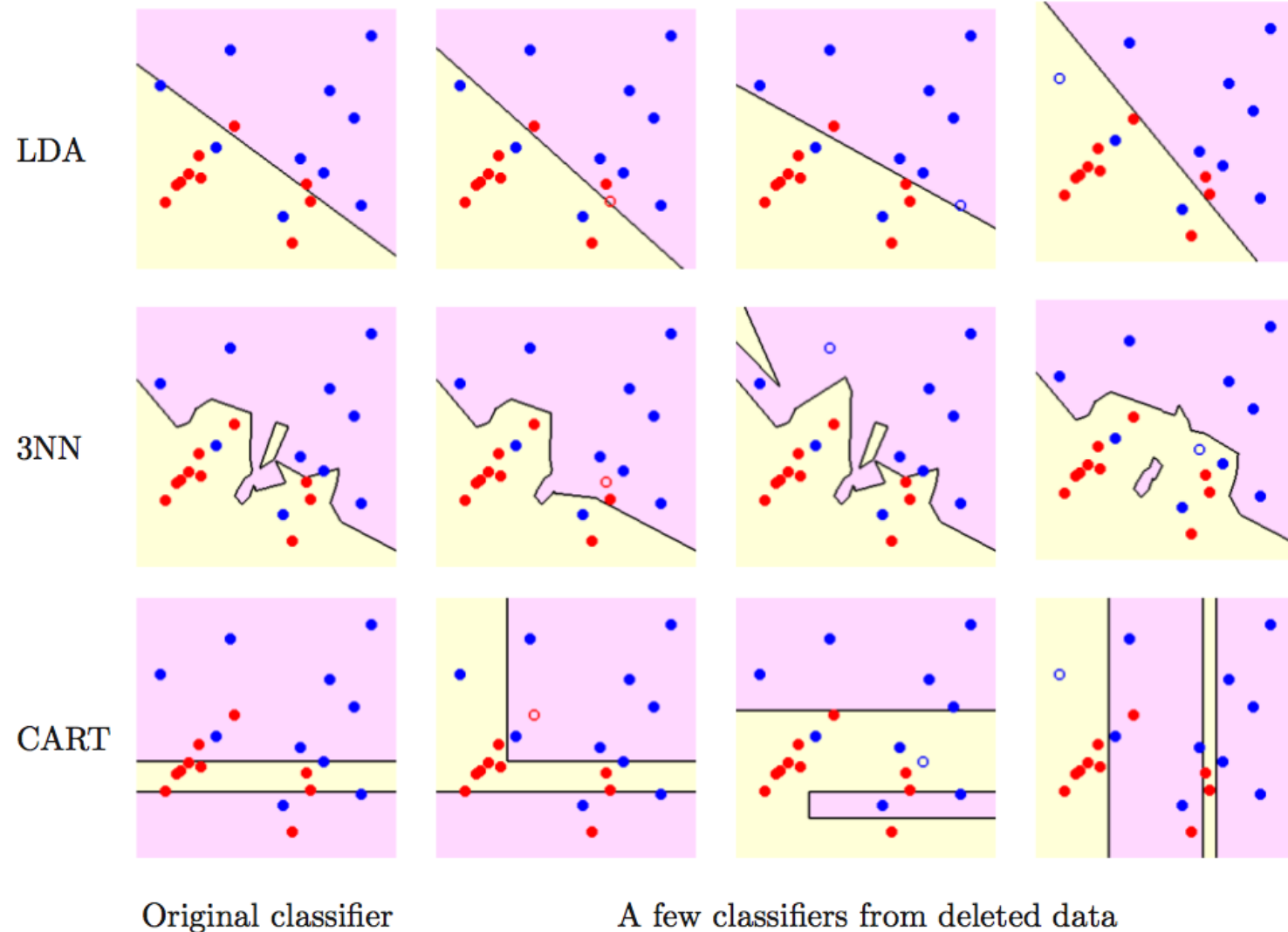
- For a symmetric classification rule,

$$\text{RMS}(\hat{\varepsilon}_n^\ell) \leq \sqrt{\frac{E[\varepsilon_{n-1}]}{n} + 6P(\Psi_n(S_n)(\mathbf{X}) \neq \Psi_{n-1}(S_{n-1})(\mathbf{X}))}$$

- **Symmetric Classification Rule:** A classification rule that is not affected by permuting the points in the training sample.
- For very large values of n , $P(\Psi_n(S_n)(\mathbf{X}) \neq \Psi_{n-1}(S_{n-1})(\mathbf{X}))$ will be small for most classification rule.
- For small n , $P(\Psi_n(S_n)(\mathbf{X}) \neq \Psi_{n-1}(S_{n-1})(\mathbf{X}))$ becomes significant.

Cross-Validation Error Estimation

- LDA is the most stable to deletion of training data.
- CART is the most unstable one.
- 3NN moderately stable.
- $P(\Psi_n(S_n)(\mathbf{X}) \neq \Psi_{n-1}(S_{n-1})(\mathbf{X}))$ is smallest for LDA $\rightarrow$ a smaller RMS for leave-one-out.
- This instability is linked with the classification rule's overfitting potential.



Cross-Validation Error Estimation

- For a symmetric classification rule,

$$\text{RMS}(\hat{\varepsilon}_n^\ell) \leq \sqrt{\frac{E[\varepsilon_{n-1}]}{n} + 6P(\Psi_n(S_n)(\mathbf{X}) \neq \Psi_{n-1}(S_{n-1})(\mathbf{X}))}$$

- Useful to calculate the distribution-independent bounds on the $\text{RMS}(\hat{\varepsilon}_n^\ell)$ for classification rules for which the bounds of $P(\Psi_n(S_n)(\mathbf{X}) \neq \Psi_{n-1}(S_{n-1})(\mathbf{X}))$ are known.
- For the k NN classification rule with random tie breaking,
 - $\Psi_n(S_n)(\mathbf{X}) \neq \Psi_{n-1}(S_{n-1})(\mathbf{X})$ is true iff. deleted point is in the k nearest neighbours.
 - The probability of this happening is k/n (all sample points equally likely to be in k nearest neighbours).
 - Also, $E[\varepsilon_{n-1}]/n < 1/n$.
 - Thus,

$$\text{RMS}(\hat{\varepsilon}_n^\ell) \leq \sqrt{\frac{6k + 1}{n}}$$

Cross-Validation Error Estimation

- k -fold CV relies on average of the surrogate classifiers designed on samples of size $n - n/k$
- It is more appropriate to consider k -fold CV as an estimator of $E[\varepsilon_{n-n/k}]$ than an estimator of $E[\varepsilon_n]$.
- If both CV and true error variance is small, it is reasonable that:

$$\begin{aligned}\hat{\varepsilon}_n &\approx E[\hat{\varepsilon}_n] \\ E[\hat{\varepsilon}_n] &= E[\varepsilon_{n-n/k}] \\ E[\varepsilon_{n-n/k}] &\approx E[\varepsilon_n] \\ E[\varepsilon_n] &\approx \varepsilon_n\end{aligned}$$

- However, under small sample size, variance is large and $\hat{\varepsilon}_n \approx E[\hat{\varepsilon}_n]$ doesn't hold.
- CV cannot provide accurate error estimation in small sample settings.

Bootstrap Error Estimation Rule

- A kind of smoothened CV with reduced variance.
- Creates multiple training sets (B bootstrap samples) by randomly sampling with replacement from original data.
- Each bootstrap sample is the same size as the original dataset, but some observations may appear multiple times while others are omitted.
- Trains a classifier on each bootstrap sample and tests it on out-of-bag (OOB) samples not included in the resamples.
- Average error across all OOB predictions giving the bootstrap estimate.
- A randomized error estimation rule.
- Different variations:
 - Standard Bootstrap: Uses random sampling with replacement
 - Balanced Bootstrap: Ensures each original observation appears approximately the same number of times across all bootstrap samples combined
 - Leave-One-Out Bootstrap: For each observation, averages predictions from only those bootstrap samples where the observation was not selected

Thank You